

The Modern Computer Vision Pipeline

Technical Research Report · Assignment 2

Abstract: Computer vision has moved from hand-engineered filters to end-to-end deep-learning systems that see, understand, and act on the visual world. This report walks through the seven stages of the modern vision pipeline: image acquisition, image processing, object detection, classification, segmentation, tracking, and deployment, explaining the role of each stage, its dominant algorithms, and how the stages combine in production systems such as autonomous vehicles, medical imaging, and automated retail.

1. Introduction

A computer-vision (CV) system converts raw photons into decisions: a car that brakes for a pedestrian, a scanner that flags a tumour, a checkout that charges the right shopper. Since AlexNet won the 2012 ImageNet challenge [1], deep neural networks have replaced most hand-crafted features, yet the surrounding engineering pipeline has remained remarkably stable. Figure 1 shows that pipeline: images are captured (§2), cleaned and standardised (§3), interpreted by one or more inference tasks (§4–6), linked over time (§7), and shipped into production (§8), with a monitoring loop feeding real-world data back into training.

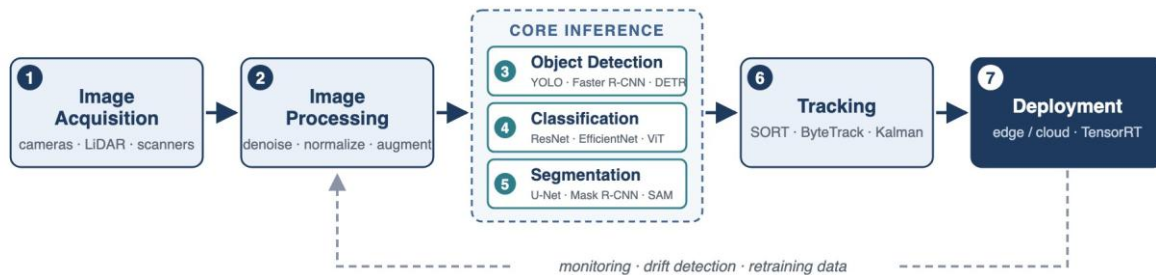


Figure 1. The end-to-end computer-vision pipeline. Core inference tasks run singly or in combination; deployment feeds monitoring data back into training (the “data flywheel”).

2. Image Acquisition

Everything downstream is limited by what the sensor captures. Acquisition covers the choice of sensor (CMOS/CCD image sensors, but also LiDAR, radar, thermal, X-ray, and MRI), the optics, resolution, frame rate, dynamic range and critically lighting. Poor acquisition cannot be fully repaired later: motion blur, glare, or under-exposure destroy information permanently, so serious systems engineer the capture environment as carefully as the model. *Real-world examples:* Waymo’s robotaxis fuse 29 cameras with LiDAR and radar for 360° coverage; factory inspection lines pair line-scan cameras with synchronised strobe lighting; smartphone night modes merge bursts of frames at capture time.

3. Image Processing

Raw sensor output is noisy and inconsistent, so classical image processing standardises it before any learning happens [2]. Typical operations include demosaicing and colour-space conversion; denoising with Gaussian, median, or bilateral filters; contrast enhancement (histogram equalisation, CLAHE); geometric correction of lens distortion; and resizing plus intensity normalisation into the exact input format the network expects. At training time, augmentation (random crops, flips, colour jitter) synthesises extra variety and reduces over-fitting. *Real-world examples:* retinal-screening pipelines apply CLAHE to fundus photographs so blood vessels stay visible across different cameras and clinics; every smartphone runs a hardware ISP performing these steps in milliseconds.

4. Object Detection

Detection answers *what* is in the image and *where*, returning a bounding box, class label, and confidence score for every object (Figure 2). Two families dominate: two-stage detectors such as Faster R-CNN, which first propose regions and then classify them for maximum accuracy [3], and one-stage detectors such as YOLO and SSD, which predict boxes in a single pass at realtime speed [4]. Transformer-based DETR removes hand-designed anchors and non-maximum suppression entirely [5].

Accuracy is scored with intersection-over-union (IoU) and mean average precision (mAP). *Real-world examples:* Amazon’s Just Walk Out stores detect products as shoppers pick them up; driver-assistance systems from Mobileye and Tesla detect vehicles and pedestrians dozens of times per second.

5. Classification

Classification assigns a single label to an image or a cropped region the oldest and most mature pipeline task. Convolutional networks pre-trained on ImageNet remain the workhorse backbones: ResNet’s skip connections made very deep networks trainable [6], EfficientNet balanced accuracy against compute, and Vision Transformers (ViT) now match or exceed CNNs when data is plentiful [7]. In practice engineers rarely train from scratch; they fine-tune a pre-trained backbone on a small taskspecific dataset (transfer learning). *Real-world examples:* grading diabetic retinopathy from retinal images at clinician-level accuracy; classifying solder-joint defects on electronics production lines; content moderation on social platforms.



Figure 2. Three core inference tasks on the same scene: a global label, a localised box, and a pixel-exact mask.

6. Segmentation

Segmentation assigns a class to every pixel, producing exact object outlines rather than coarse boxes (Figure 2). Semantic segmentation labels pixels by category; instance segmentation additionally separates individual objects; panoptic segmentation unifies both. Encoder–decoder architectures such as U-Net dominate medical imaging [8], Mask R-CNN extends detection with per-object masks [9], and the promptable Segment Anything Model (SAM) generalises zero-shot to unseen objects [10]. *Real-world examples:* radiotherapy planning software contours tumours and organs-at-risk on CT scans; video-call background blur segments the speaker in real time; precision-agriculture drones map weeds for targeted spraying.

7. Tracking

Video adds time. Tracking links per-frame detections into persistent identities and trajectories the tracking-by-detection paradigm. A Kalman filter predicts where each object will move; the Hungarian algorithm matches predictions to fresh detections (SORT); DeepSORT adds appearance re-identification features so identities survive occlusion [11]; ByteTrack also recovers low-confidence detections to reduce identity switches [12]. Quality is measured with MOTA and IDF1. *Real-world examples:* Hawk-Eye tracks balls and players for broadcast sport and officiating; traffic authorities count and classify vehicles per lane; retailers analyse anonymised shopper flow through stores.

8. Deployment

A model only creates value once it runs inside a product under real latency, power, and cost budgets. Trained models are exported to an interchange format (ONNX) and compiled with runtimes such as TensorRT or Core ML; INT8 quantisation, pruning, and knowledge distillation shrink them further often 4–10× with little accuracy loss. Edge deployment gives low latency and privacy; cloud deployment gives elastic compute; many products combine both. Deployment is also where MLOps begins: live accuracy monitoring, data-drift detection, and a retraining loop close the cycle shown in Figure 1. *Real-world examples:* NVIDIA Jetson modules run detection on inspection drones; Tesla executes its vision stack on a custom in-car chip; AWS Rekognition and Google Cloud Vision serve vision models as cloud APIs. Table 1 summarises the full pipeline.

Table 1. The pipeline at a glance: representative methods and deployed systems.

Stage	Representative methods	Deployed real-world system
Image acquisition	CMOS cameras, LiDAR, thermal, MRI	Waymo multi-sensor robotaxi rig
Image processing	Denoising, CLAHE, normalisation, augmentation	Smartphone ISP / night mode
Object detection	YOLO, Faster R-CNN, DETR	Amazon Just Walk Out checkout
Classification	ResNet, EfficientNet, ViT	Diabetic-retinopathy screening
Segmentation	U-Net, Mask R-CNN, SAM	Radiotherapy tumour contouring
Tracking	SORT, DeepSORT, ByteTrack	Hawk-Eye sports analytics
Deployment	ONNX, TensorRT, INT8 quantisation	Jetson-powered inspection drones

9. Conclusion

The modern CV pipeline is a chain of specialised stages, and its overall quality is set by its weakest link: a state-of-the-art detector cannot compensate for poor lighting, and a perfect model is useless if it misses its latency budget. Transformer backbones, promptable foundation models such as SAM, and mature MLOps tooling are making individual stages more general but the pipeline structure itself, from acquisition through deployment, remains the blueprint of every production vision system.

References

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